AI-Powered YouTube Course Mentor using Retrieval-Augmented Generation (RAG)

Domain: Education Technology | Artificial Intelligence | NLP | RAG

Problem Description

Students learning from long YouTube courses often struggle to find specific topics, revise concepts, take notes, and get answers to doubts. Traditional video platforms do not provide intelligent learning assistance based on course content.

Objective

Develop an AI-powered YouTube Course Mentor that extracts transcripts, builds a searchable knowledge base, answers questions, generates notes and quizzes, and improves learning efficiency.

Proposed Solution

The system uses Retrieval-Augmented Generation (RAG). Video transcripts are extracted, chunked, converted into embeddings, stored in a vector database, and retrieved to answer student questions accurately.

System Workflow

1. Collect YouTube transcripts
2. Clean and chunk text
3. Generate embeddings
4. Store in ChromaDB
5. Retrieve relevant chunks
6. Generate answers using an LLM
7. Provide notes and quizzes

Expected Outcome

Interactive AI tutor, faster learning, easier revision, automated note generation, quiz generation, and personalized educational assistance.

Technologies Used

React.js, FastAPI, LangChain, ChromaDB, Sentence Transformers, YouTube Transcript API, Gemini API / Llama 3.

Conclusion

The project transforms YouTube courses into an intelligent learning assistant using RAG and LLMs, making online learning more interactive and effective.